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Volume 04 — Large Language Models, RAG, Multimodal AI, and Ethics

Lectures L28-L36 · Weeks 10-12

L28: Transformer Architecture and Self-Attention

Learning objectives

- Describe the full transformer encoder block.
- Implement multi-head self-attention.

Transformer block

flowchart TB

```
IN["Input embeddings + positional encoding"] --> MHA["Multi-Head Self-Attention"]
MHA --> ADD1["Add & Norm"]
IN --> ADD1
ADD1 --> FFN["Feed-Forward Network"]
FFN --> ADD2["Add & Norm"]
ADD1 --> ADD2
ADD2 --> OUT["Output"]
```

Multi-head attention

For head i :

$$\text{head}_i = \text{Attention}(XW_i^Q, XW_i^K, XW_i^V)$$

$$\text{MultiHead}(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

Feed-forward sublayer

$$\text{FFN}(x) = \max(0, xW_1 + b_1) W_2 + b_2$$

Applied position-wise (identical network at each position, different parameters from attention).

Layer normalization and residuals

$$x' = \text{LayerNorm}(x + \text{Sublayer}(x))$$

Complexity

Self-attention has $O(n^2d)$ time and memory for sequence length n , compared to $O(nd^2)$ for RNNs—but with full parallelism over n .

Decoder-only vs. encoder-decoder

Architecture	Used in	Masking
Encoder-only	BERT	Bidirectional
Decoder-only	GPT	Causal (left-to-right)
Encoder-decoder	T5, BART	Cross-attention

L29: Tokenization, Embeddings, and Positional Encoding

Learning objectives

- Compare tokenization strategies (BPE, WordPiece, SentencePiece).
- Explain positional encoding schemes.

Tokenization

Text is split into **tokens**—subword units that balance vocabulary size and coverage:

Method	Used by	Idea
BPE	GPT-2/3/4	Merge frequent byte pairs
WordPiece	BERT	Likelihood-based merges
SentencePiece	T5, Llama	Language-agnostic; no pre-tokenization

Embeddings

Token IDs are mapped to vectors via a learned embedding matrix $E \in \mathbb{R}^{|V| \times d}$:

$$x_i = E[\text{token_id}_i]$$

Sinusoidal positional encoding (original Transformer)

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d}), \quad PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d})$$

Learned positional embeddings

Modern models (GPT, BERT) use learned position embeddings. **RoPE** (Rotary Position Embedding) encodes relative positions in attention dot products and scales to long contexts.

L30: Prompt Engineering and In-Context Learning

Learning objectives

- Apply prompting strategies for LLM tasks.
- Explain in-context learning without gradient updates.

Prompting strategies

Strategy	Description
Zero-shot	Task description only
Few-shot	Include exemplar input-output pairs
Chain-of-thought	Ask model to reason step by step
System prompt	Set role, constraints, format

In-context learning

LLMs conditioned on demonstration examples $\{(x_i, y_i)\}$ perform tasks without parameter updates:

$$P(y \mid x, \{(x_i, y_i)\}_{i=1}^k)$$

This emerges from pre-training on diverse text but is not equivalent to true learning—performance is sensitive to example selection and ordering.

Scaling laws

Model performance improves predictably with compute, data, and parameters (Kaplan et al., 2020; Hoffmann et al., 2022—Chinchilla optimal allocation).

L31: Retrieval-Augmented Generation Architecture

Learning objectives

- Design a RAG pipeline combining retrieval and generation.
- Understand when RAG outperforms fine-tuning alone.

RAG architecture

flowchart LR

```
Q["User query"] --> EMB["Query encoder"]
EMB --> RET["Retriever"]
KB["Knowledge base\n(documents)"] --> RET
RET --> TOPK["Top-k passages"]
TOPK --> CTX["Augmented context"]
Q --> CTX
CTX --> LLM["LLM generator"]
LLM --> ANS["Grounded answer"]
```

Formal formulation (Lewis et al., 2020)

Given query x , retrieve documents $z \in \mathcal{Z}$ and generate:

$$p(y \mid x) = \sum_{z \in \mathcal{Z}} p_{\eta}(z \mid x) p_{\theta}(y \mid x, z)$$

where p_{η} is the retriever and p_{θ} is the generator (LLM).

Retrieval scoring

Dense retrieval with bi-encoder:

$$\text{score}(q, d) = \text{sim}(E_q(q), E_d(d)) = \frac{E_q(q)^{\top} E_d(d)}{\|E_q(q)\| \|E_d(d)\|}$$

Advantages over fine-tuning

- **Updatable knowledge** — swap document index without retraining.
- **Attribution** — cite retrieved passages.
- **Reduced hallucination** — grounded in retrieved evidence.

L32: Vector Databases and Document Chunking

Learning objectives

- Implement document chunking strategies for RAG.
- Select and use vector databases.

Chunking strategies

Strategy	Chunk size	Overlap	Use case
Fixed-size	256–512 tokens	10–20%	General documents
Semantic	Variable	None	Structured content
Recursive	Hierarchical	Parent-child	Long reports

Embedding pipeline

1. Chunk documents $\rightarrow \{c_1, \dots, c_M\}$.
2. Embed each chunk: $v_i = E_d(c_i)$.
3. Store $(v_i, \text{metadata}_i)$ in vector database.

Vector databases

System	Features
FAISS	In-memory, fast similarity search
Pinecone	Managed, scalable
Chroma	Lightweight, local
pgvector	PostgreSQL extension

Approximate nearest neighbor (ANN)

For large corpora, exact search is too slow. ANN algorithms (HNSW, IVF) trade accuracy for speed:

$$\mathcal{Z}_k = \text{ANN}_k(E_q(q), \{v_i\}_{i=1}^M)$$

L33: Fine-Tuning with LoRA and Parameter-Efficient Methods

Learning objectives

- Apply LoRA for efficient fine-tuning.
- Compare PEFT methods.

Full fine-tuning cost

Updating all parameters of a 7B model requires storing optimizer states for 7B+ parameters—prohibitive for most practitioners.

LoRA (Low-Rank Adaptation)

Freeze pretrained weights W_0 and add a low-rank update:

$$W = W_0 + \Delta W = W_0 + BA$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $\text{rank } r \ll \min(d, k)$.

Train only A and B —typically $< 1\%$ of original parameters.

Other PEFT methods

Method	Mechanism
Prefix tuning	Learnable prefix tokens prepended to input
Adapter layers	Small bottleneck layers inserted in transformer
QLoRA	LoRA + 4-bit quantized base model

When to use RAG vs. fine-tuning

Criterion	RAG	Fine-tuning
Dynamic knowledge	□	□
Style/format control	Limited	□
Latency	Higher (retrieval step)	Lower
Infrastructure	Vector DB required	GPU for training

L34: Multimodal AI: Vision-Language Models

Learning objectives

- Describe CLIP and its contrastive training.
- Survey multimodal LLM architectures.

CLIP (Contrastive Language-Image Pre-training)

Train image encoder f_I and text encoder f_T with contrastive loss:

$$\mathcal{L} = -\frac{1}{N} \sum_i \log \frac{\exp(\text{sim}(f_I(x_i), f_T(t_i))/\tau)}{\sum_j \exp(\text{sim}(f_I(x_i), f_T(t_j))/\tau)}$$

Enables zero-shot image classification via text prompts.

Multimodal LLMs

Model	Approach
LLaVA	Project vision features into LLM token space
GPT-4V	Unified multimodal transformer
Flamingo	Cross-attention to visual features
Stable Diffusion	Text-conditioned image generation (diffusion)

Unified architectures

Modern trend: treat images, audio, and text as token sequences in a single transformer, with modality-specific encoders projecting to a shared embedding space.

L35: LLM Evaluation, Hallucinations, and Safety

Learning objectives

- Apply standard LLM benchmarks.
- Identify and mitigate hallucinations.

Evaluation benchmarks

Benchmark	Measures
MMLU	Multi-domain knowledge (57 subjects)
HellaSwag	Commonsense reasoning
HumanEval	Code generation
TruthfulQA	Factual accuracy
MT-Bench	Multi-turn conversation quality

Hallucinations

Hallucination = model generates plausible but factually incorrect content.

Type	Example
Intrinsic	Contradicts source document
Extrinsic	Invents facts not in any source

Mitigation

1. **RAG** — ground responses in retrieved documents.
2. **Constitutional AI** — train with principle-based feedback.
3. **Uncertainty estimation** — calibrate confidence; abstain when uncertain.
4. **Human-in-the-loop** — review high-stakes outputs.

Safety alignment

- **RLHF** (Reinforcement Learning from Human Feedback): train reward model on human preferences, optimize policy with PPO.
 - **DPO** (Direct Preference Optimization): bypass reward model; optimize directly on preference pairs.
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L36: Bias, Fairness, and Ethics in Generative AI

Learning objectives

- Identify sources of bias in generative models.
- Apply ethical frameworks for AI deployment.

Sources of bias

1. **Training data** — reflects societal biases, underrepresentation.
2. **Objective function** — likelihood maximization doesn't encode fairness.
3. **Deployment context** — model used beyond intended scope.

Fairness metrics

Metric	Definition
Demographic parity	Equal positive rates across groups
Equalized odds	Equal TPR and FPR across groups
Calibration	Predicted probabilities match outcomes

Environmental impact

Training large models consumes significant energy. Inference at scale adds ongoing cost. Consider model size vs. task requirements.

Ethical principles

1. **Transparency** — disclose AI-generated content.
2. **Accountability** — human oversight for consequential decisions.
3. **Privacy** — protect training data; avoid memorization attacks.
4. **Accessibility** — consider who benefits and who is harmed.

Regulatory landscape

- EU AI Act (risk-based classification)
- NIST AI Risk Management Framework
- India's Digital Personal Data Protection Act

Course conclusion

This course traced generative AI from first principles—probability, neural networks, and optimization—through VAEs, GANs, diffusion models, transformers, and LLMs. The field evolves rapidly; the mathematical foundations developed here provide durable tools for understanding new architectures as they emerge.

Volume 04 summary

Lecture	Core concept
L28	Transformer architecture
L29	Tokenization, positional encoding
L30	Prompt engineering, in-context learning
L31	RAG architecture
L32	Vector DBs, chunking
L33	LoRA, PEFT
L34	Multimodal AI, CLIP
L35	Evaluation, hallucinations
L36	Bias, fairness, ethics

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